Machine Learning 2 (APM_50441_EP/DSAIB - 2026-2027). PC2

Exercise 1 (Link Estimation/Prediction)

Assume we observe a sample $((X_i, Y_i))_{i=1}^n$ where $Y_i \in \{-1, 1\}$ and $X_i \in \mathbb{R}^d$ are independent random variables following the same law as a generic pair (X, Y). We want to predict Y from X using a predictor $f : \mathbb{R}^d \to \{-1, 1\}$ and we measure the quality of our predictor using the $0/1$ loss:

$$\ell^{0/1}(y, y') = \mathbf{1}_{y \neq y'}.$$

The optimal Bayes classifier is given by

$$f^\star(X) = \operatorname{sign}(2\eta(X) - 1)$$

where $\eta(X) = \mathbb{P}(Y = 1|X)$.

Assume we have an estimate $\widehat{\eta}(X)$ of $\eta(X)$, i.e. an estimate of the conditional law of $Y|X$. We define the plug-in classifier as

$$\widehat{f} = \operatorname{sign}(2\widehat{\eta}(X) - 1).$$

We represent the conditional law of $Y|X$ and its estimate by the probability vectors

$$p(X) = (\eta(X), 1 - \eta(X)) \quad \text{and} \quad \widehat{p}(X) = (\widehat{\eta}(X), 1 - \widehat{\eta}(X)),$$

so that $\|p(X) - \widehat{p}(X)\|_1 = 2|\eta(X) - \widehat{\eta}(X)|$.

We want to prove that

$$\begin{aligned}\mathbb{E}\left[\ell^{0/1}(Y, \widehat{f}(X))\right] - \mathbb{E}\left[\ell^{0/1}(Y, f^\star(X))\right] &\leq \mathbb{E}[\|\widehat{p}(X) - p(X)\|_1] \\ &\leq \left(\mathbb{E}[2\,\mathrm{KL}(p(X), \widehat{p}(X))]\right)^{1/2}.\end{aligned}$$

1. Prove that for any predictor f,

$$\mathbb{E}\left[\ell^{0/1}(Y, f(X))\right] = \mathbb{E}_X\left[(1 - \eta(X)) + (2\eta(X) - 1)\mathbf{1}_{f(X)=-1}\right]$$

2. Deduce that

$$\begin{aligned}\mathbb{E}\left[\ell^{0/1}(Y, \widehat{f}(X))\right] - \mathbb{E}\left[\ell^{0/1}(Y, f^\star(X))\right] \\ \leq 2\mathbb{E}_X[|\eta(X) - \widehat{\eta}(X)|] = \mathbb{E}_X[\|p(X) - \widehat{p}(X)\|_1]\end{aligned}$$

3. Finish the proof using $\|P - Q\|_1 \leq \sqrt{2\,\mathrm{KL}(P, Q)}$.

Exercise 2 (Link surrogate loss/prediction)

We want to show that the sign of the minimizer of

$$\mathbb{E}\left[\bar{\ell}(Y, f(\underline{X}))\right] = \mathbb{E}[l(Y f(\underline{X}))],$$

where l is a convex non increasing function such that $l(0) = 1$, l is differentiable at 0 and $l'(0) < 0$, is the Bayes classifier $f^\star = \operatorname{sign}(2\mathbb{P}(Y = 1|\underline{X}) - 1)$. This covers the logistic loss $l(x) = \log_2(1 + e^{-x})$ of the lecture, for which $l'(0) = -1/(2\log 2)$, the exponential loss e^{-x} and the hinge loss $\max(0, 1 - x)$.

1. Write $\mathbb{E}\left[\bar{\ell}(Y, f(\underline{X}))\right]$ as a function $H(f, \eta(\underline{X}))$ where $\eta(\underline{X}) = \mathbb{P}(Y = 1|\underline{X})$.

2. Prove that, for $\eta \neq 1/2$, any minimizer $\tilde{f}$ of $H(\cdot, \eta)$ has the same sign as $2\eta - 1$.

3. Conclude

4. *What does the minimizer tell about η?* For $\eta \in (0,1)$, compute the minimizer $\tilde{f}(\eta)$ of $H(\cdot,\eta)$ for the logistic loss $l(x) = \log_2(1+e^{-x})$, the exponential loss $l(x) = e^{-x}$, the hinge loss $l(x) = \max(0, 1-x)$ and the squared loss $l(x) = (1-x)^2$ (which is not non increasing, but is used for classification too). For which losses can η be recovered from $\tilde{f}$? Relate this to Lab 2, where these losses are minimized numerically.

Exercise 3 (Linear Regression and Design)

Assume we observe some points $((X_i, Y_i))_{i=1}^n$ with $X_i \in \mathbb{R}^d$ and $Y_i \in \mathbb{R}$. Regressing those points means estimating the regression function $f^\star(X) = \mathbb{E}[Y|X]$, at least at the X_i. We denote by f a generic predictor built from the observations (it is a function of the sample), and we keep the notation $\hat{f}$ for the empirical risk minimizer, the estimator obtained by training (see below). There are two classical problems that can be studied:

- **Fixed Design** We assume that the X_i are fixed and want to estimate $f^\star$ only at the observed points X_i. We measure the quality of f as the average error on a replication set $((X_i, Y_i'))_{i=1}^n$:

$$\mathsf{Err}_F(f) = \mathbb{E}_{(Y_i)_{i=1}^n}\left[\mathbb{E}_{(Y_i')_{i=1}^n}\left[\frac{1}{n}\sum_{i=1}^n \ell(Y_i', f(X_i))\right]\right]$$

  where the X_i are the same and $Y_i'|X_i$ has the same law as $Y_i|X_i$.

- **Random Design** We assume that the X_i are i.i.d. and we want to estimate $f^\star$ everywhere. We measure the quality as the average error on a new observation (X', Y') having the same law as (X_i, Y_i):

$$\mathsf{Err}_R(f) = \mathbb{E}_{((X_i,Y_i))_{i=1}^n}\left[\mathbb{E}_{(X',Y')}[\ell(Y', f(X'))]\right].$$

Linear least squares regression is classically studied in the fixed design setting, while the random design setting corresponds to the statistical learning setting. In this exercise, we are going to look at the performance of the linear least squares estimator in those two settings.

Following Rosset and Tibshirani[1], we introduce an intermediate setting:

- **Repeated Fixed Design** We assume that the X_i are random, possibly of different laws, and want to estimate $f^\star$ only at the observed points X_i. We measure the quality by the Fixed Design error averaged over all possible designs:

$$\mathsf{Err}_{RF}(f) = \mathbb{E}_{((X_i,Y_i))_{i=1}^n}\left[\mathbb{E}_{(Y_i')_{i=1}^n}\left[\frac{1}{n}\sum_{i=1}^n \ell(Y_i', f(X_i))\right]\right],$$

as well as the empirical estimate of the quality:

$$\mathsf{Err}_{Emp}(f) = \frac{1}{n}\sum_{i=1}^n \ell(Y_i, f(X_i)).$$

We assume that all the samples are independent conditionally to the X_i, that the X_i are fixed in the Fixed design setting, independent in the Repeated Fixed design setting and i.i.d. in the Random design one. Except for the warm-up, we will assume a homoscedastic setting in which $Y = f^\star(X) + \epsilon$ with ϵ a centered random noise of variance σ^2 independent of X, and that the loss is the square one.

1. Warm-up

   (a) Verify that $\mathbb{E}_{(X_i)_{i=1}^n}[\mathsf{Err}_F(f)] = \mathsf{Err}_{RF}(f)$ and, if the (X_i, Y_i) are i.i.d.,

$$\mathsf{Err}_R(f) = \mathbb{E}_{((X_i,Y_i))_{i=1}^n}\left[\mathbb{E}_{((X_i',Y_i'))_{i=1}^n}\left[\frac{1}{n}\sum_{i=1}^n \ell(Y_i', f(X_i'))\right]\right]$$

   where the (X_i', Y_i') are independent copies of the (X_i, Y_i).

[1] S. Rosset and R. J. Tibshirani, *From Fixed-X to Random-X Regression: Bias-Variance Decompositions, Covariance Penalties, and Prediction Error Estimation*, Journal of the American Statistical Association, 115(529), 2020.

(b) Verify that if $\hat{f} \in \operatorname{argmin}_{g \in \mathcal{G}} \sum_{i=1}^n \ell(Y_i, g(X_i))$ for a fixed class $\mathcal{G}$ of predictors then $\mathbb{E}_{(Y_i)_{i=1}^n}\left[\mathsf{Err}_{Emp}(\hat{f})\right] \leq \mathsf{Err}_F(\hat{f})$ and $\mathbb{E}_{((X_i,Y_i))_{i=1}^n}\left[\mathsf{Err}_{Emp}(\hat{f})\right] \leq \mathsf{Err}_{RF}(\hat{f})$.

2. We will now assume that ℓ is the ℓ^2 loss: $\ell(Y, f(X)) = |Y - f(X)|^2$.

   (a) Prove that in the fixed design setting:

$$\mathsf{Err}_F(f) = \sigma^2 + \frac{1}{n}\sum_{i=1}^n (f^\star(X_i) - \mathbb{E}_{(Y_i)_{i=1}^n}[f(X_i)])^2 + \frac{1}{n}\sum_{i=1}^n \mathbb{V}\mathrm{ar}_{(Y_i)_{i=1}^n}[f(X_i)]$$

   (b) Show that if the X_i are i.i.d., and if f does not depend on the order of the observations (as the least squares estimator), then for the repeated fixed design setting:

$$\mathsf{Err}_{RF}(f) = \sigma^2 + \mathbb{E}_{(X_i)_{i=1}^n}\left[(f^\star(X_1) - \mathbb{E}_{(Y_i)_{i=1}^n}[f(X_1)|(X_i)_{i=1}^n])^2\right] + \mathbb{E}_{(X_i)_{i=1}^n}\left[\mathbb{V}\mathrm{ar}_{(Y_i)_{i=1}^n}[f(X_1)|(X_i)_{i=1}^n]\right]$$

   (c) Finally prove that for the random design setting

$$\mathsf{Err}_R(f) = \sigma^2 + \mathbb{E}_{(X_i)_{i=1}^n, X'}\left[(f^\star(X') - \mathbb{E}_{(Y_i)_{i=1}^n}[f(X')|(X_i)_{i=1}^n])^2\right] + \mathbb{E}_{(X_i)_{i=1}^n, X'}\left[\mathbb{V}\mathrm{ar}_{(Y_i)_{i=1}^n}[f(X')|(X_i)_{i=1}^n]\right]$$

3. We define the optimism $\mathsf{Opt}_F(f)$ (respectively $\mathsf{Opt}_{RF}(f)$) as the difference between $\mathsf{Err}_F(f)$ (respectively $\mathsf{Err}_{RF}(f)$) and the corresponding expectation of $\mathsf{Err}_{Emp}(f)$.

   (a) Justify the name *optimism* when the estimator is the empirical risk minimizer $\hat{f}$.

   (b) Verify that in the fixed design setting:

$$\mathsf{Opt}_F(f) = \mathsf{Err}_F(f) - \mathbb{E}_{(Y_i)_{i=1}^n}[\mathsf{Err}_{Emp}(f)] = \mathbb{E}_{(Y_i)_{i=1}^n, (Y_i')_{i=1}^n}\left[\frac{1}{n}\sum_{i=1}^n (Y_i' - f(X_i))^2 - \frac{1}{n}\sum_{i=1}^n (Y_i - f(X_i))^2\right]$$

   (c) Prove that

$$\mathsf{Opt}_F(f) = \frac{2}{n}\sum_{i=1}^n \mathbb{C}\mathrm{ov}_{(Y_i)_{i=1}^n}[Y_i, f(X_i)]$$

   (d) Deduce that in the repeated fixed design setting

$$\mathsf{Opt}_{RF}(f) = \frac{2}{n}\sum_{i=1}^n \mathbb{E}_{(X_i)_{i=1}^n}\left[\mathbb{C}\mathrm{ov}_{(Y_i)_{i=1}^n}[Y_i, f(X_i)|(X_i)_{i=1}^n]\right]$$

4. We consider now the random design and define $\mathsf{Opt}_R(f) = \mathsf{Err}_R(f) - \mathbb{E}_{((X_i,Y_i))_{i=1}^n}[\mathsf{Err}_{Emp}(f)]$ in the same way.

   (a) Verify that

$$\mathsf{Err}_R(f) = \mathsf{Err}_{RF}(f) + \underbrace{\mathbb{E}_{(X_i)_{i=1}^n, X'}\left[(f^\star(X') - \mathbb{E}_{(Y_i)}[f(X')|(X_i)_{i=1}^n])^2\right] - \mathbb{E}_{(X_i)}\left[(f^\star(X_1) - \mathbb{E}_{(Y_i)}[f(X_1)|(X_i)])^2\right]}_{\Delta_B}$$
$$+ \underbrace{\mathbb{E}_{(X_i)_{i=1}^n, X'}\left[\mathbb{V}\mathrm{ar}_{(Y_i)}[f(X')|(X_i)_{i=1}^n]\right] - \mathbb{E}_{(X_i)_{i=1}^n}\left[\mathbb{V}\mathrm{ar}_{(Y_i)}[f(X_1)|(X_i)_{i=1}^n]\right]}_{\Delta_V}$$

   (b) Deduce that

$$\mathsf{Opt}_R(f) = \mathsf{Opt}_{RF}(f) + \Delta_B + \Delta_V$$

   (c) Why is it reasonable to expect that both Δ_B and Δ_V are nonnegative?

5. We consider now the linear model $f_\beta(x) = \langle x, \beta\rangle$ and the least squares estimator $\hat{f} = f_{\widehat{\beta}}$, with $\widehat{\beta}$ minimizing the empirical loss. We denote by $\mathbb{X}$ the $n \times d$ design matrix, whose rows are the ${X_i}^\top$, and by $\mathbf{Y}$ the vector of the Y_i.

(a) Prove that, assuming that $\mathbb{X}^\top \mathbb{X}$ is an invertible $d \times d$ matrix, the estimate $\widehat{\beta}$ is given by

$$\widehat{\beta} = \left(\mathbb{X}^\top \mathbb{X}\right)^{-1} \mathbb{X}^\top \mathbf{Y}$$

so that $\hat{f}(X_i) = (\mathrm{Proj}_{\mathrm{Im}\,\mathbb{X}} \mathbf{Y})_i$, where $\mathrm{Proj}_{\mathrm{Im}\,\mathbb{X}} = \mathbb{X}\left(\mathbb{X}^\top \mathbb{X}\right)^{-1} \mathbb{X}^\top$ is the orthogonal projection onto the column space of $\mathbb{X}$.

(b) Deduce that $\mathsf{Opt}_F(\hat{f}) = \mathsf{Opt}_{RF}(\hat{f}) = 2\sigma^2 \frac{d}{n}$.

(c) Prove that $\Delta_B \geq 0$ for $f = \hat{f}$.

(d) Prove that $\Delta_V \geq 0$ for $f = \hat{f}$ (one can use that $\mathbb{E}\left[\left(\mathbb{X}^\top \mathbb{X}\right)^{-1}\right] - \left(\mathbb{E}\left[\mathbb{X}^\top \mathbb{X}\right]\right)^{-1}$ is positive semi-definite).

Exercise 4 (Linear discriminant analysis is a linear classifier)

In the fully generative approach, the classes $Y \in \{0, 1\}$ have prior probabilities π_0, π_1 and the conditional laws of $X \in \mathbb{R}^d$ are Gaussian, $X|Y = k \sim \mathcal{N}(\mu_k, \Sigma_k)$.

1. Compute the log-odds $\log \frac{\mathbb{P}(Y=1|X=x)}{\mathbb{P}(Y=0|X=x)}$.
2. Show that when $\Sigma_0 = \Sigma_1 = \Sigma$ (LDA), the log-odds is an affine function $\beta_0 + \beta^\top x$ of x, and give β and β_0. Deduce that the LDA and the logistic regression share the same model for $\mathbb{P}(Y = 1|X = x)$. Why are their estimates different?

Exercise 5 (Effective number of parameters of the ridge regression)

We consider the ridge regression $\widehat{\beta}_\lambda = \mathrm{argmin}_\beta \|\mathbf{Y} - \mathbb{X}\beta\|^2 + \lambda\|\beta\|^2$ with $\lambda > 0$, where $\mathbb{X}$ is a fixed $n \times d$ design matrix with singular value decomposition $\mathbb{X} = UDV^\top$ (D diagonal with entries $d_1 \geqslant \cdots \geqslant d_r > 0$).

1. Show that $\widehat{\beta}_\lambda = (\mathbb{X}^\top \mathbb{X} + \lambda I)^{-1} \mathbb{X}^\top \mathbf{Y}$ and that the fitted values are $\widehat{\mathbf{Y}} = S_\lambda \mathbf{Y}$ with $S_\lambda = \sum_{j=1}^r \frac{d_j^2}{d_j^2 + \lambda} u_j u_j^\top$.
2. Using Exercise 7.5 of ESL (PC1), deduce the effective number of parameters $\mathrm{df}(\lambda) = \mathrm{tr}(S_\lambda)$, and its limits when $\lambda \to 0$ and $\lambda \to \infty$. How can it be used to compare the regularization paths of Lab 2?

Exercise 6 (Support vector machine: slack variables and hinge loss)

The soft-margin SVM solves

$$\min_{\beta, \beta_0, \xi} \frac{1}{2}\|\beta\|^2 + C \sum_{i=1}^n \xi_i \quad \text{subject to} \quad Y_i(\beta^\top X_i + \beta_0) \geqslant 1 - \xi_i,\ \xi_i \geqslant 0\,,$$

with labels $Y_i \in \{-1, 1\}$ and $C > 0$.

1. Show that, at the optimum, $\xi_i = \max(0, 1 - Y_i(\beta^\top X_i + \beta_0))$, and deduce that the SVM minimizes

$$\frac{1}{n} \sum_{i=1}^n \max\left(0, 1 - Y_i(\beta^\top X_i + \beta_0)\right) + \lambda\|\beta\|^2$$

for a value of λ to be given as a function of C and n.

2. Which observations have $\xi_i > 0$? Which observations influence the solution? Relate this to the number of support vectors reported in Lab 2.

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